Exp.No.9 Implementation of Subnetting in CISCO PACKET TRACER simulator

28/9/26

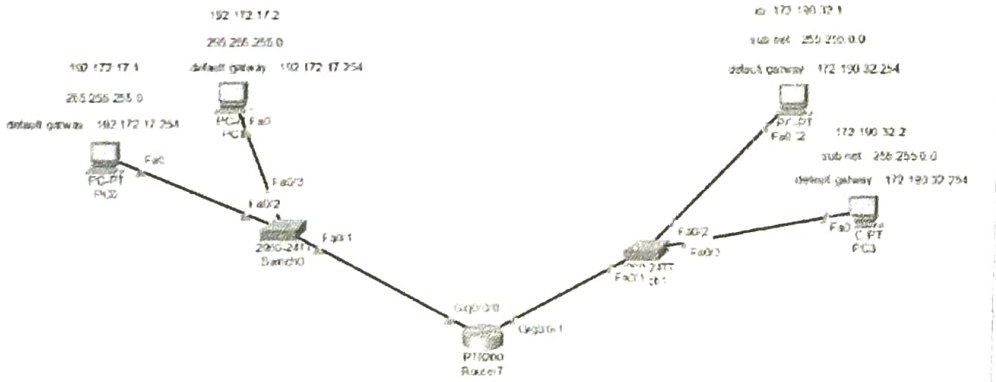
Aim:

To implement of subnetting in CISCO PACKET TRACER simulator.

Procedure

* Open Cisco Packet tracer and create a new network topology
* Add the required router, switches and PCs to the workspace.
* Connect the device using appropriate copper cables
* Use the network 192.168.1.0/24 and divide it into 27 networks
* Configure the router interfaces with the required IP address and subnet mask 255.255.255.224
* Use enable and configure terminal in the router CLI
* Enter the interface configuration terminal in the router assign the IP using:
    i) ip address < IP address > 255.255.255.224
    ii) no shut down.
* Configure the required switch ports in access mode
* Assign each PC its IP address, subnet mask and default gateway
* Repeat the configuration for all required subnets
* Use the ping command to test connectivity between PCs and the router
* If ping is successful, the network is configured correctly.

```
Router>
*Sep 24, 11:17:27.1717: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0/1, changed state to up
Router>
Router>
Router>
Router>
Router>
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.172.17.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)# ip address 172.190.32.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
```



Result.

Thus the implementation of subnetting in Cisco packet Tracker Simulator is successful